



CSCE 633-700: Machine Learning

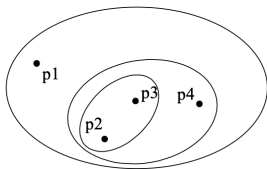
Clustering

Overview

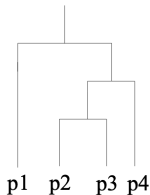
- Clustering overview
- Partitional clustering
 - K-means clustering
 - Gaussian Mixture Models (GMM)
- Hierarchical clustering

Hierarchical clustering

- Produces a set of nested clusters organized as a hierarchical tree
- Visualized as a dendrogram
 - A tree like diagram that records the sequences of merges or splits



Traditional Hierarchical Clustering



Traditional Dendrogram

Hierarchical clustering

Advantages of hierarchical clustering

- Do not have to pre-determine number of clusters
 - Any desired number of clusters can be obtained by 'cutting' the dendrogram at the proper level
- Resulting clusters may correspond to meaningful taxonomies
 - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction)

Hierarchical clustering

Types of hierarchical clustering

- Agglomerative
 - Start with each sample as individual cluster
 - Merge the closest pair of clusters each time until only one cluster left
- Divisive
 - Start with one, all-inclusive cluster
 - Split a cluster each time until each cluster contains a point

Agglomerative hierarchical clustering

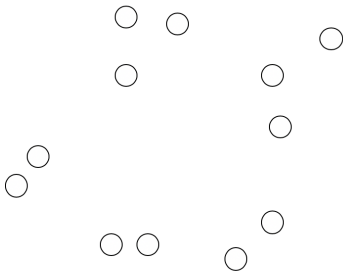
- **Step 0:** Compute the proximity matrix
- **Step 1:** Let each data sample be a cluster
- **Step 2: Repeat:**
 - Merge the two closest clusters
 - Update the proximity matrix

Until only a single cluster remains

Key operation is the computation of the **proximity** of two clusters → different approaches for defining distance between clusters

Agglomerative hierarchical clustering

Initialization: Start with each sample being a cluster



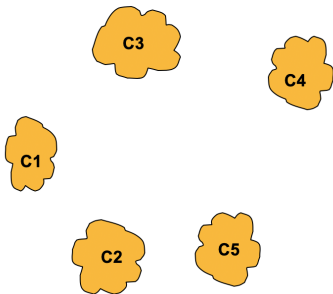
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix



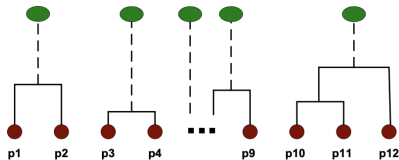
Agglomerative hierarchical clustering

After some steps: we have some clusters



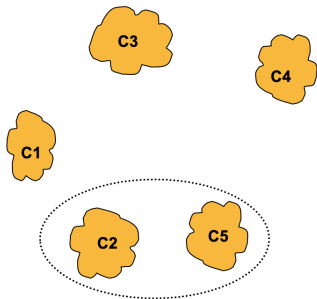
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



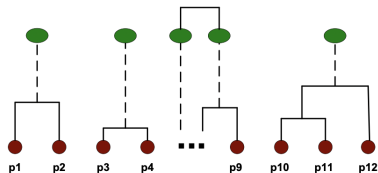
Agglomerative hierarchical clustering

We want to merge the two closest clusters (C2 and C5) and update the proximity matrix



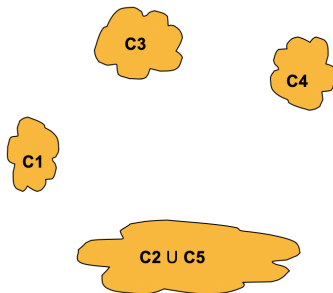
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



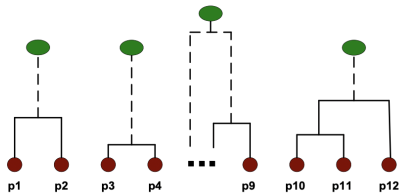
Agglomerative hierarchical clustering

How do we update the proximity matrix?



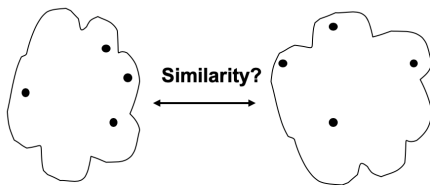
		C2 U C5	C3	C4
C1		?		
C2 U C5	?	?	?	?
C3		?		
C4		?		

Proximity Matrix



Agglomerative hierarchical clustering

How to define inter-cluster similarity?

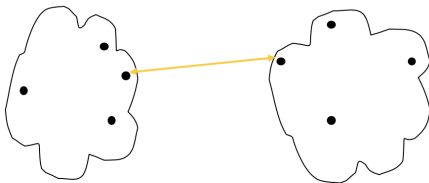


	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						

- min, max, group average, distance between centroids

Agglomerative hierarchical clustering

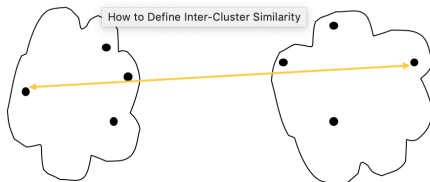
Distance between the closest samples (min)



	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						

Agglomerative hierarchical clustering

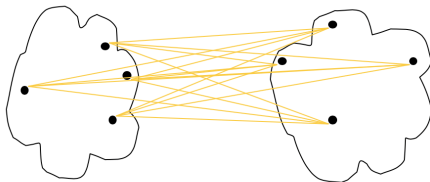
Distance between the furthest samples (max)



	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						

Agglomerative hierarchical clustering

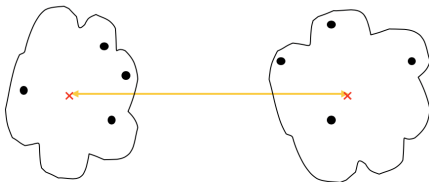
Average pairwise distance between samples (group average)



	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						

Agglomerative hierarchical clustering

Distance between centroids



	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						